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Getting Started

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What is Combinatorial Game?

Optional

Syllabus

Games without Chance: Combinatorial Game Theory: Syllabus

Prerequisite Knowledge

This is a university level Mathematics course. While there are no specific Mathematics prerequisites, it is highly recommended that students have taken rigorous college level Mathematics courses (AP calculus counts as such).

Course Textbook or Online Resources

Although the class is designed to be self-contained, students wanting to expand their knowledge beyond what we can cover in a one-quarter class can find a much more extensive coverage of this topic in the 4 volume work "Winning Ways for your Mathematical Plays" by Elwyn R. Berlekamp, John H. Conway and Richard K. Guy.

Grading Policy

There will be one quiz each week for weeks 2 through 6 (no assignment week 1 or 7). A quiz will consist of a few problems, usually 4 or 5. Each question is worth 1 point each with no partial credit. All answers should be submitted to the Quizzes tab on the left by the deadline of the quiz. The deadline for a quiz will be Sunday at 11:59 PM EST (the end of the current week). You will have until Monday at 11:59 PM EST to submit the quiz without a penalty, and until Wednesday at 11:59 PM EST to submit it with a 10% penalty per day late. A detailed solutions guide will be uploaded after the deadline. To pass the course, an overall score of 60% will be needed. To pass the course with distinction, an overall score of 85% will be needed. **EDIT: There will be a bonus quiz during Week 7 in case you missed a quiz or missed a lot of questions for some reason. This bonus quiz will be 5 questions, each question worth 4 points. However many points you get out of this 20 will be added to your final score (so if you have a 50 and get every bonus question right, your score will be 50+20=70). In order to pass with distinction, however, you must have above an 85 WITHOUT the bonus quiz.**

Communication

Please communicate with the instructor and staff via the online forums. Because there are so many students, it is not possible for us to respond individually to every question.

Netiquette

Written language will be primary means of communication. As such, there can be miscommunication as there is no intonation in these written communications. Please be positive, supportive and constructive in your comments and forum postings.

Expectations

Participants are expected to:

- Watch all lecture videos every week
- Complete all quizzes
- Abide by the standards of academic honesty – plagiarism and any form of cheating will not be tolerated and will result in the removal of the participant from the course
- Seek help if needed

Plagiarism

Plagiarism is considered a serious offense. Please make sure that all materials that you submit in any assignments, projects or quizzes are your own. If you copy and paste or copy word for word from the Internet or any other source, that is considered plagiarism. You will be immediately removed from the class if you plagiarize.

Course Outline

Week 1

- Let's play a game: Students will learn what a combinatorial game is, and play simple games
 - * Introduction to the two players: Left and Right. Ending rules.
 - # Some new games: Hakenbush, push, shove, run over.
- What is NOT a combinatorial game: dice, cards, etc.
 - * Formal definition of games in terms of left and right options.
- Problems and Solutions

Week 2

- Playing several games at once, adding games, the negative of a game. Student will be able to add simple games and analyze them.
- Definition of the 0 game, the negative of a game.
 - * Examples from Hakenbush
 - * Problem and solution
- The Tweedledee- Tweedledum principle.
 - * Proof that $G - G = 0$. Problem and solution.
- Homework/Quiz

Week 3

- Comparing games. Students will determine the outcome of simple sums of games using inequalities.
 - * Definition of game inequalities in terms of who does better. Problem and solution.
 - * Using best play to decide inequalities. Problem and solution.
 - * The games 1,2,3,...,-1,-2,... Examples from Ski jumps
 - # Problem and solution.
 - * The game $\frac{1}{2} = \{0|1\}$. Proof that $\frac{1}{2} + \frac{1}{2} = 1$
 - * Other dyadic rational number games...
 - * Games that are not numbers: $\star$, $\uparrow$, others. Problem and solution
- HW/quiz

Week 4

- Simplicity and numbers. How to play win numbers. Students will be able to determine which games are numbers and if so what numbers they are.
 - * When is a game a number?
 - * If a game is a number, then which number? How to play numbers
 - # Examples. Problem and solution
 - * Examples from the games Cut Cake and Maundy Cake.
 - # Problem and solution
- HW/quiz

Week 5

- Simplifying games: Dominating moves, reversible moves.
- Students will be able to simplify simple games.
 - * New ways of simplifying games. Dominated moves, reversible moves.
 - # Problem and solution
 - * Examples of reversible moves from Nim.
 - # Problem and solution
 - * More simple games: Up, Star, Down. Adding these.
 - # Problem and solution
 - * Examples from toads and frogs
 - # Problem and solution
- HW/quiz

Week 6

- Nim: Students will be able to play and analyze impartial games.
 - * What is nim and how to play it. Nim sums.
 - # Problem and solution
 - * Definition of impartial games Reversible moves in impartial games
 - # Problem and solution
 - * Examples of reversible moves in impartial games. The minimum excluded positive integer function.
 - # Problem and solution
 - * All impartial games are equivalent to Nim
- HW/quiz

Week 7

- Where to go from here:
 - * The small small small world: games that are infinitesimal
 - * The Big Big Big world: Games that are infinite
 - * Some open problems in combinatorial games.

Go to next item

✓ Completed

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